Experiment 6

Time Series Analysis of Website Traffic

```python
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.seasonal import seasonal_decompose

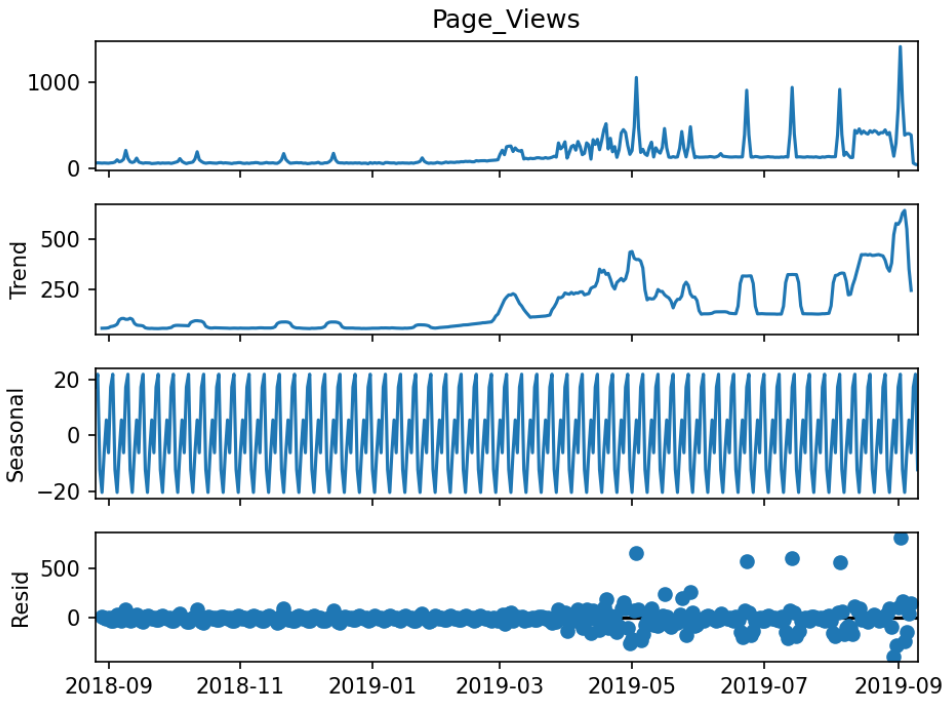
# Load dataset
data = pd.read_csv('website_traffic.csv', parse_dates=['Date'], index_col='Date')

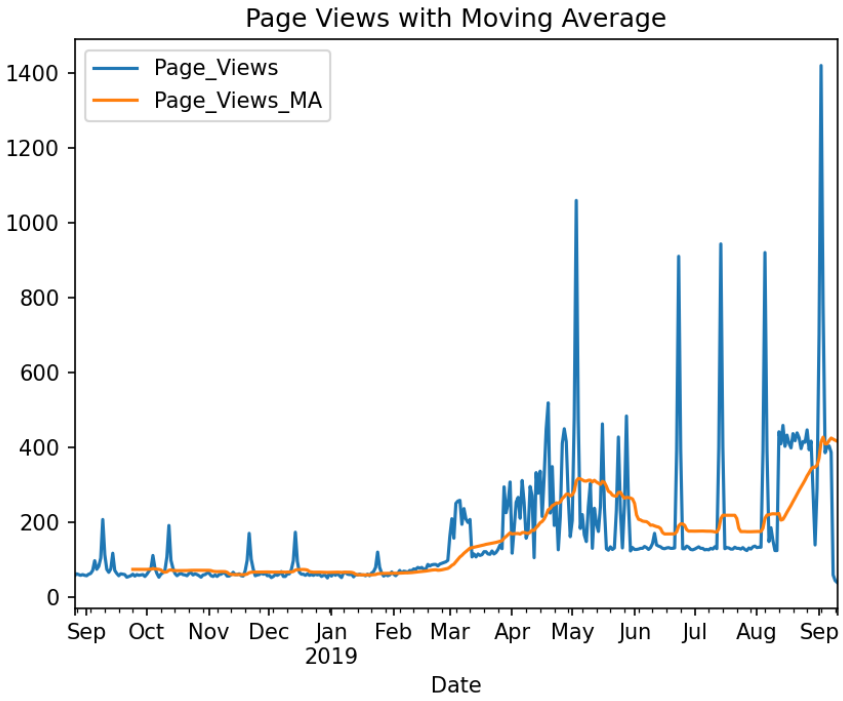
# Time Series Decomposition
decomposition = seasonal_decompose(data['Page_Views'], model='additive')
decomposition.plot()
plt.show()

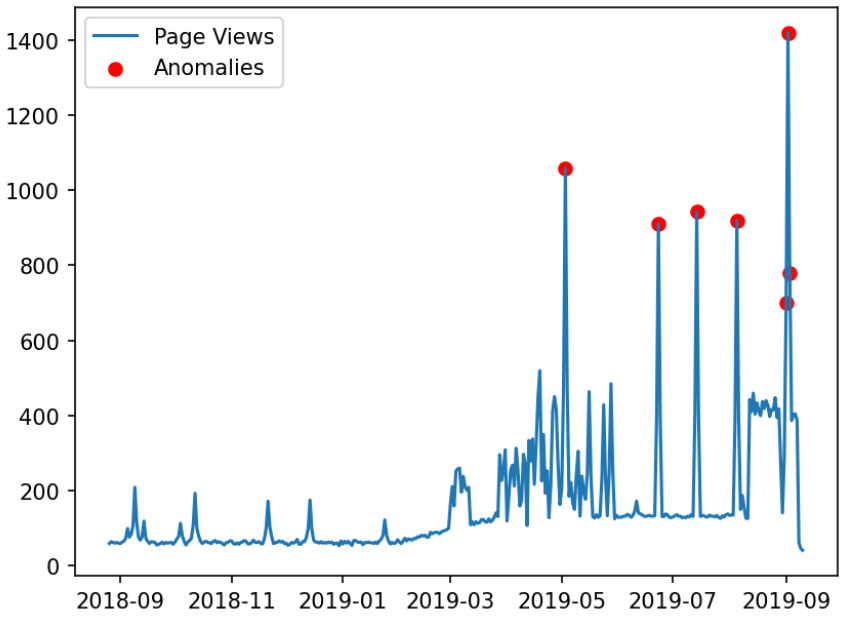
# Moving Average
data['Page_Views_MA'] = data['Page_Views'].rolling(window=30).mean()
data[['Page_Views', 'Page_Views_MA']].plot()
plt.title('Page Views with Moving Average')
plt.show()

# Anomaly Detection
data['Page_Views_Z'] = (data['Page_Views'] - data['Page_Views'].mean()) / data['Page_Views']
    .std()
anomalies = data[data['Page_Views_Z'].abs() > 3]
plt.plot(data.index, data['Page_Views'], label='Page Views')
plt.scatter(anomalies.index, anomalies['Page_Views'], color='red', label='Anomalies')
plt.legend()
plt.show()
```

3.2s




Page Views with Moving Average
Page_Views
Page_Views_MA
1400
1200
1000
800
600
400
200
0
Sep
Oct
Nov
Dec
Jan
2019
Feb
Mar
Apr
May
Jun
Jul
Aug
Sep
Date


Page Views
Anomalies
1400
1200
1000
800
600
400
200
0
2018-09
2018-11
2019-01
2019-03
2019-05
2019-07
2019-09